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Who may read our messages?

Privacy vs. state surveillance of communication
— with Chat Control as a case study

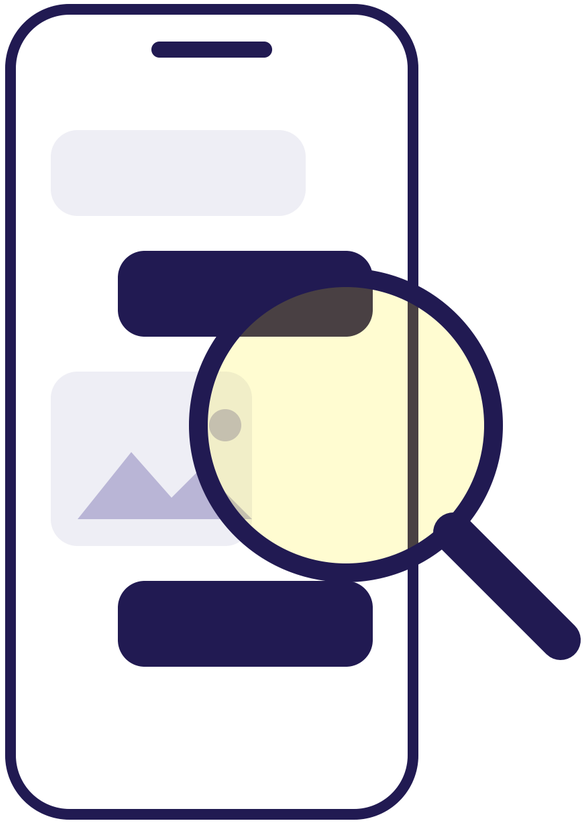
[Group members]
Dilemmas in Cyber Security E26 · Aalborg University Copenhagen



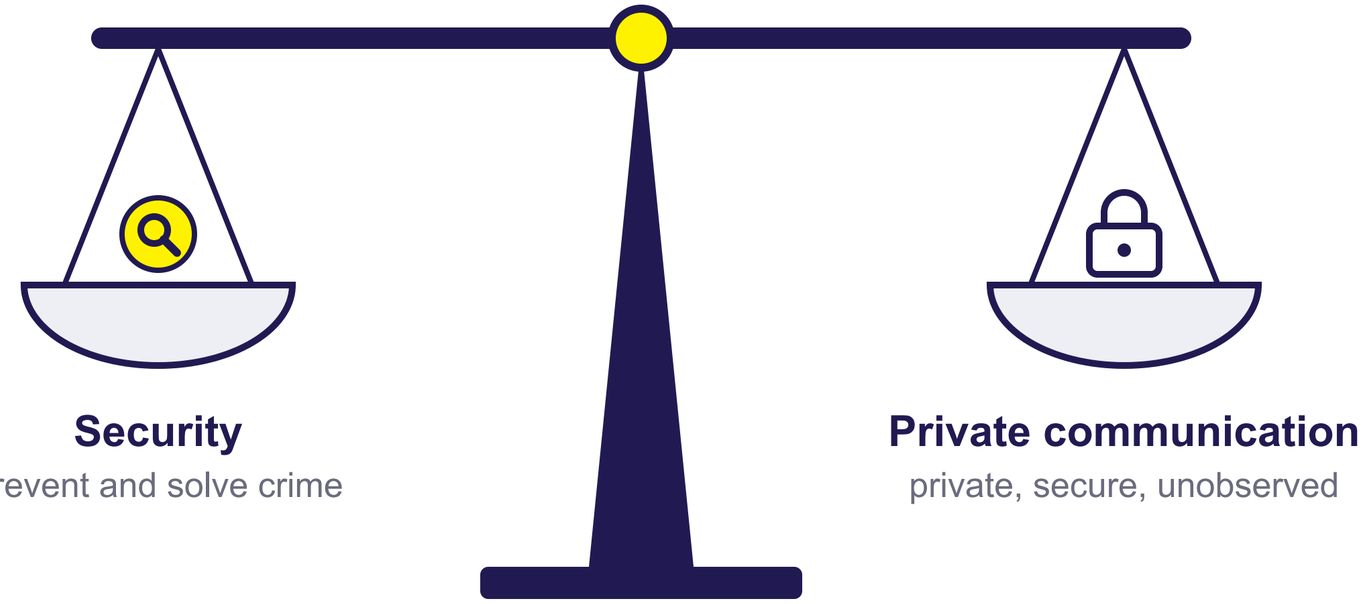
Who should be able to read your messages?

Most people would say: me, the person I write to, and in some cases the police with a court order.

Where that line is drawn has been argued about for more than 30 years.



Two legitimate goals on one scale



Security
prevent and solve crime

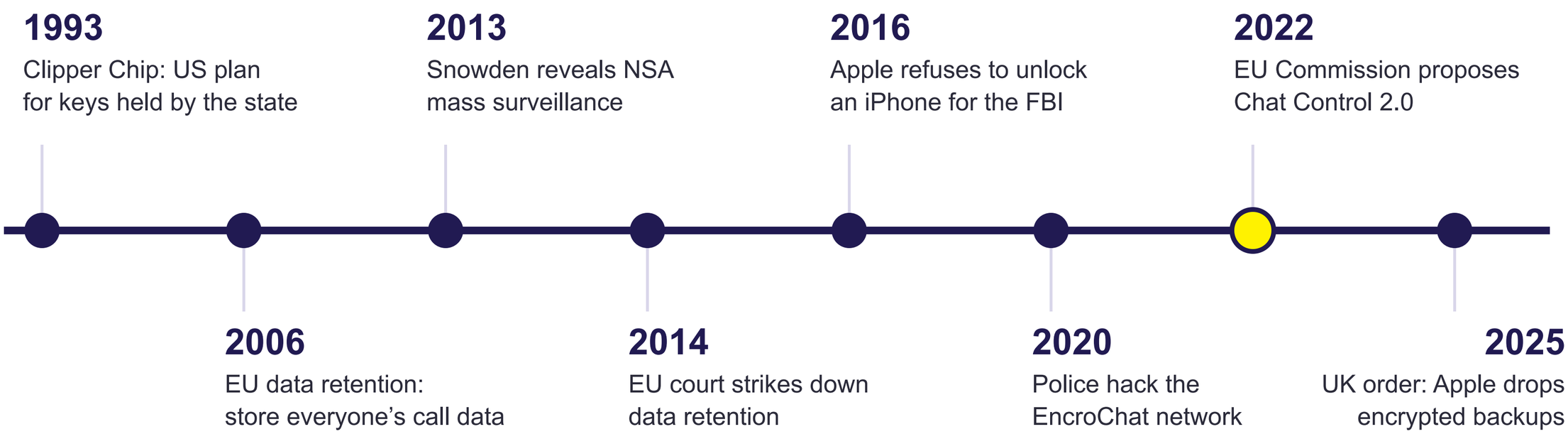
Private communication
private, secure, unobserved

The state has a duty to protect people, for example children, from serious crime.

People have a right to private communication (EU Charter, Art. 7 and 8).

More access for the state means less confidentiality, and the other way round. Every option gives something up.

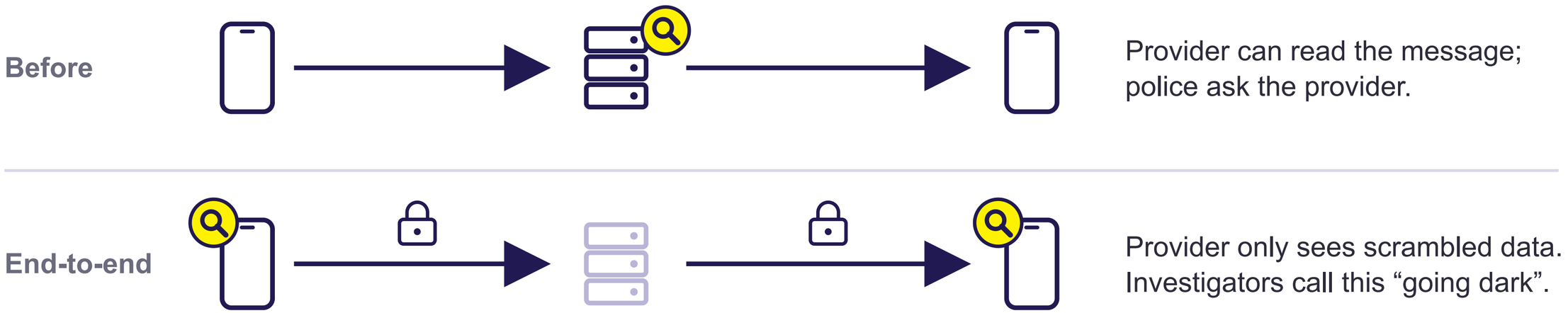
The same conflict comes back with every new technology



Four ways the state can listen

	Targeted: one suspect	Mass: everyone
Content	**Wiretap, state trojan, spyware** What a suspect says or writes. Germany 2024: 16,868 wiretap orders. Needs a judge's order.	**Bulk interception, scanning** Everyone's content is searched. NSA programmes (Snowden, 2013); **Chat Control 2.0** (proposed 2022).
Metadata	**Call records of a suspect** Who, when, where — for one person. e.g. "quick freeze" of data after a crime.	**Data retention** Everyone's connection data is stored in advance. EU law struck down 2014; Germany debates IP storage (2026).

Encryption moved the point of access to our phones



The state’s three options today

1 · Hack the device
state trojans, spyware

2 · Scan on the device
client-side scanning (Chat Control)

3 · Weaken encryption
backdoors (UK vs. Apple, 2025)

Part 2

Why privacy matters

Arguments against more surveillance

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“I’ve got nothing to hide”

The most common argument for surveillance, and three problems with it (Solove, 2007)



Small facts add up

Harmless data points, combined, show health, beliefs, relationships and routines.



It is about power

Whoever holds the data decides how it is used. The person concerned cannot check or correct it.

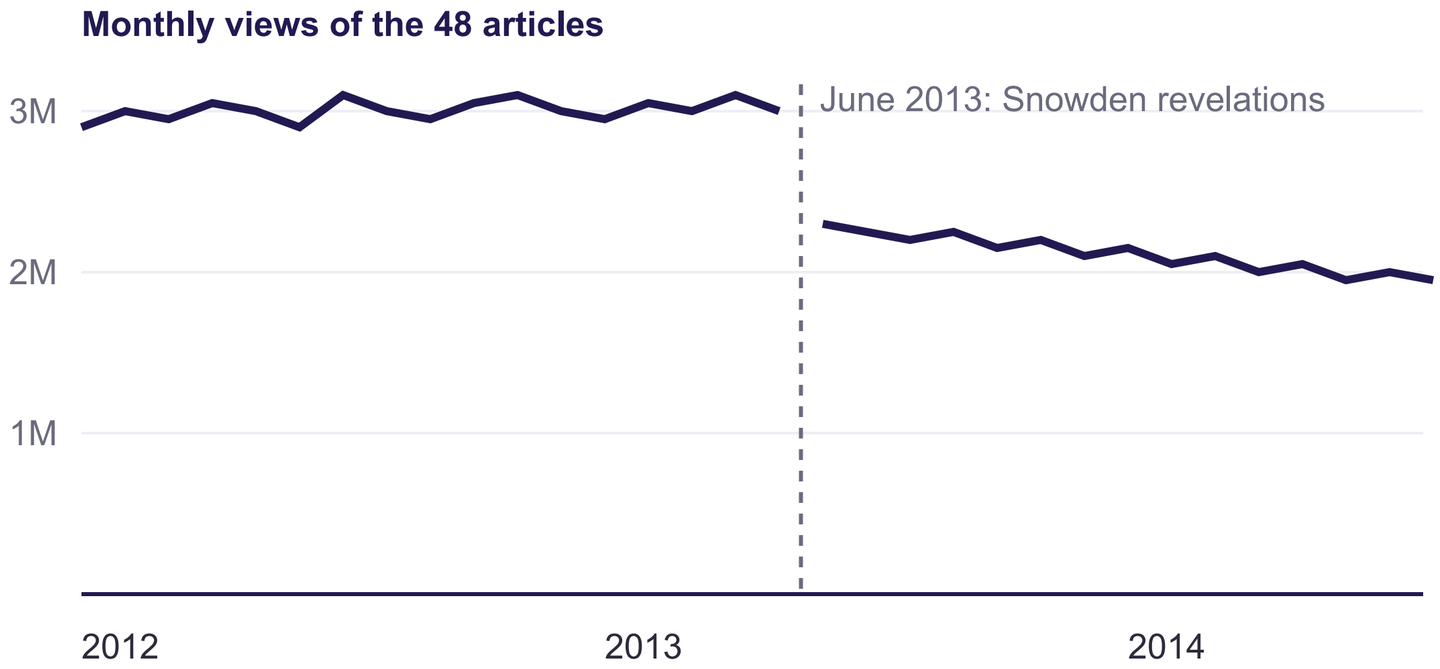


Rules change

What is harmless today can become a problem under a future government or in another country.

Solove, D. J. (2007). “I’ve Got Nothing to Hide” and Other Misunderstandings of Privacy. San Diego Law Review, 44, 745.

People change their behaviour when they feel watched



−20%

monthly views of 48 Wikipedia articles on terrorism-related topics after the Snowden revelations in June 2013.

Views kept falling in the following months. Researchers call this a **chilling effect**.

Penney, J. (2016). Chilling Effects: Online Surveillance and Wikipedia Use. Berkeley Technology Law Journal, 31, 117. Chart simplified.

Access built for the police can be used by others

Greece, 2004–05

The Athens Affair

Unknown attackers used the legal wiretap function in Vodafone Greece's network to listen to about **100 phones**, including the Prime Minister's.



USA and worldwide, 2024–25

Salt Typhoon

Chinese state hackers got into US telecom networks, including systems used for court-ordered wiretaps. In 2025, 13 countries reported **200+ US organisations** affected and **80+ countries** targeted.

In Dec 2024 the FBI and CISA advised Americans to use **encrypted messengers**.

Security researchers conclude that a way in that works only for authorities cannot be built (Abelson et al., 2015).

Prevelakis & Spinellis (2007), IEEE Spectrum; joint advisory of 13 countries (27 Aug 2025); Abelson et al. (2015), Keys under doormats

Surveillance tools spread beyond their original purpose



50,000

phone numbers on a leaked list of possible **Pegasus** spyware targets (2021)

~90

WhatsApp users warned in 2025 about **Paragon** spyware, incl. journalists in Italy

2025

German Constitutional Court bans state trojans for **minor offences**

Pegasus Project (Forbidden Stories / Amnesty, 2021); TechCrunch (2026) on Paragon; BVerfG 1 BvR 180/23 (2025)

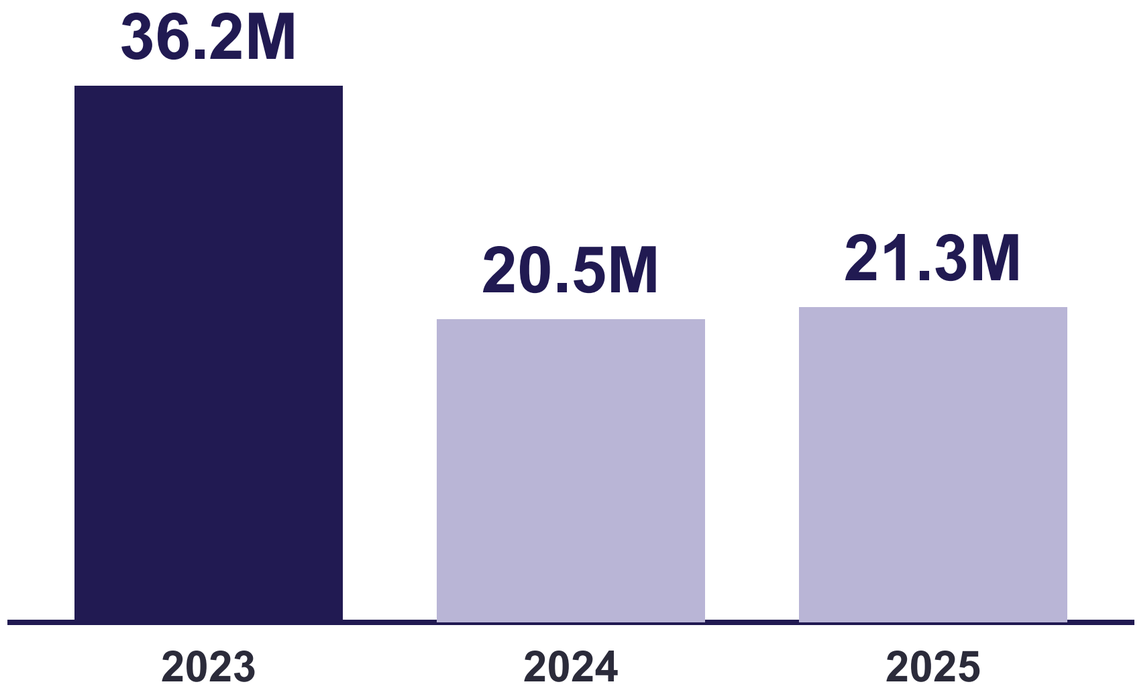
Part 3

Why the state wants access

Arguments for more surveillance

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Encryption also hides serious crime



–43%

fewer reports of child abuse material due to:

- bundling
- Messenger's end-to-end encryption

The REPORT Act (2024) was meant to increase reporting: platforms must now also report child trafficking and online enticement.

Reports to NCMEC, the US hotline most platforms report to (CyberTipline data 2023–2025; NBC News, 2025)

Targeted operations against encrypted networks

Police hacked or secretly ran encrypted phone services used mainly by criminals

EncroChat, 2020

6,558

arrests across Europe, about €900M seized. Around 60,000 users.

ANOM, 2021

800+

arrests in 16 countries. The FBI secretly ran the app on 12,000 devices.

Sky ECC, 2021

585

arrest warrants in Germany alone (until 2025). About 70,000 users monitored.

These cases show that investigators can reach encrypted communication without scanning everyone. Courts are still debating how such evidence may be used (CJEU, C-670/22, 2024).

Europol / Eurojust (2023); FBI & Australian Federal Police (2021); BKA (2025)

Part 4

Case study: Chat Control

The EU plan to scan private chats for child abuse material

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Chat Control 1.0 and 2.0

2021

1.0 · Companies may scan

- After EU extension of ePrivacy Directive to messaging apps
- The exception lets tech companies scan messages voluntarily

2022

2.0 · Companies must scan

- Authorities could order a service to scan all its users' messages, including encrypted ones
- A new EU Centre would handle reports in Europe instead of sending them to NCMEC

Client-side scanning



Why researchers call it a backdoor

The encryption stays intact. Its promise does not.

2021

NeuralHash by Apple

- Photos were compared on the iPhone with a list of known abuse images
- Apple cancelled the project in 2022
- Its lesson: privacy and scanning exclude each other, and the scanner becomes a new target

Abelson et al. (2024), Bugs in our pockets

Four years of negotiations, no agreement yet

May 2022
Commission: scanning mandatory

Nov 2023
Parliament: only suspects, judge’s order

Nov 2025
Member states: voluntary scanning

Sep 2026
6th round: no deal

Dec 2026?
next round reported

Agreed

Risk assessment for platforms, removal and blocking orders, the EU Centre.

Still open

Scanning orders for private chats, encrypted services, age verification.

EP negotiating team (29 Jun 2026); Council doc. 11956/26 (10 Sep 2026); netzpolitik.org (2 Oct 2026). Status: 27 Oct 2026.

Who stands where

← Scan everyone

Scan no one →



EU Commission

2022 proposal:
all chats, incl. encrypted

+ child-protection NGOs, Europol

Member states

voluntary scanning by companies

+ Meta, Google, Microsoft

EU Parliament

only suspects, with a judge's order

+ Germany, data-protection authorities

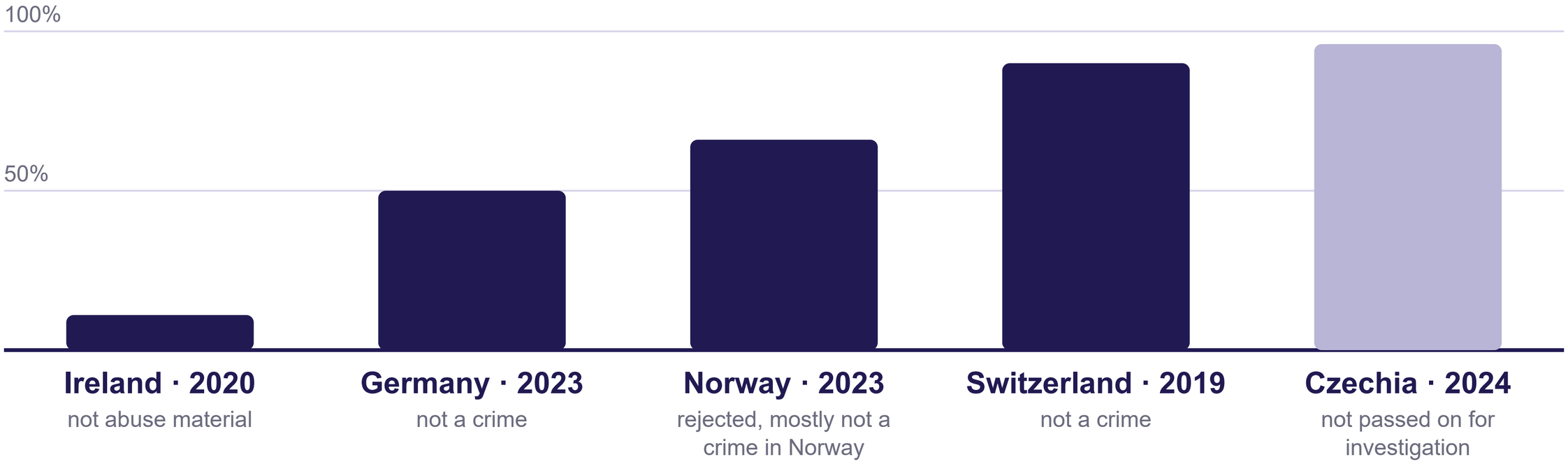
Signal & scientists

no scanning of private chats

~500 researchers; Signal would leave the EU

Many alarms concern legal content

Share of reports to police that did not lead to a case



Part 5

Weighing the dilemma

One framework from the course

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Five questions to weigh the dilemma

Principlist framework for cybersecurity ethics — Formosa, Wilson & Richards (2021), Lecture 2



Beneficence

Does it **do good**?



Non-maleficence

Does it **cause harm**?



Autonomy

Does it respect people's **freedom**?



Justice

Is it **fair** to everyone?

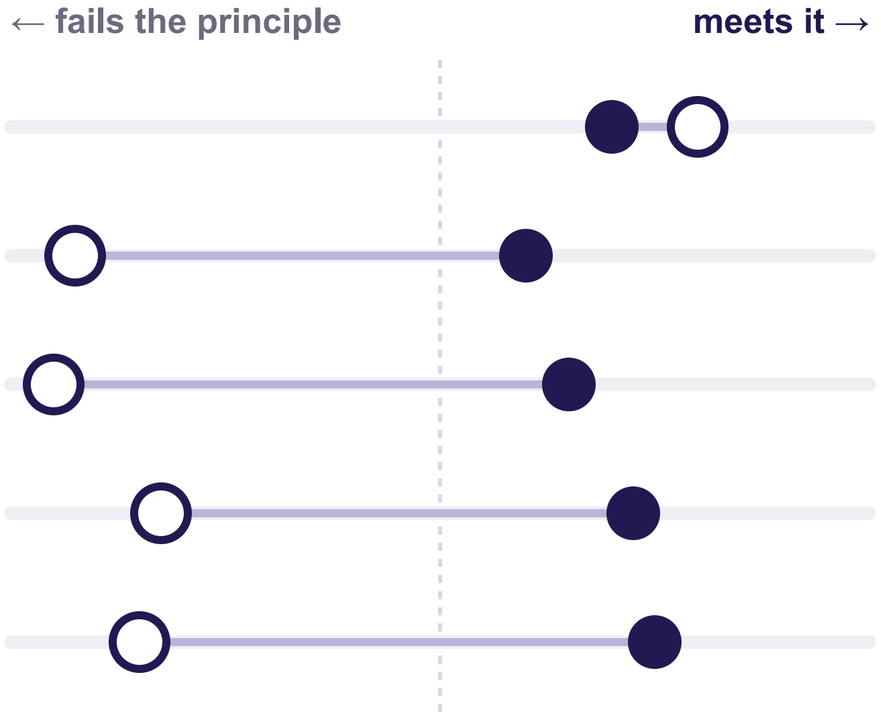


Explicability

Is it **transparent** and accountable?

We use the five questions to compare two approaches: scanning everyone (Chat Control 2.0) and targeted measures against suspects.

Mass scanning vs. targeted measures



both find offenders; mass scanning finds more

mass: weaker security, innocent people flagged

mass: everyone watched, chilling effect

mass: every user treated as a suspect

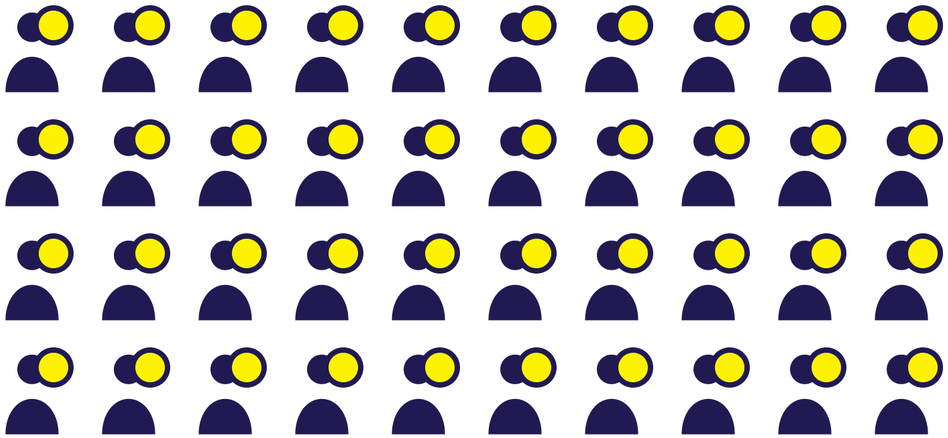
targeted: judge’s order, documented

○ Mass scanning (Chat Control 2.0) ● Targeted measures (judge’s order)

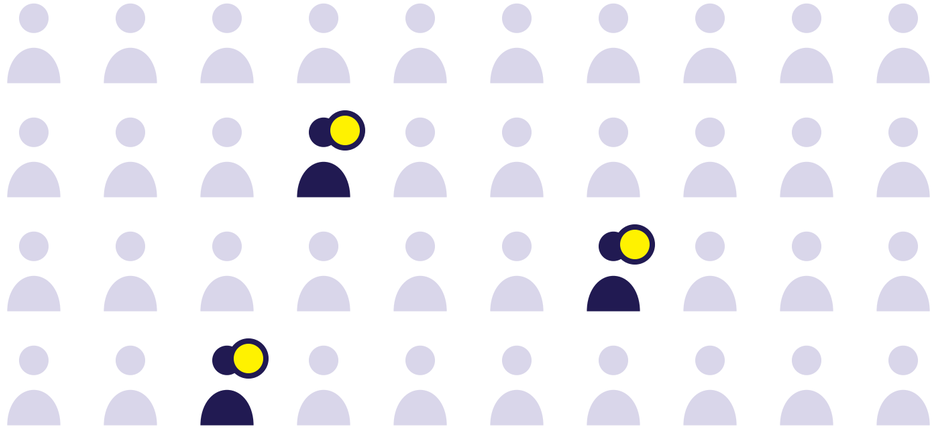
Positions show our group’s assessment, based on the evidence in this presentation.

Check everyone, or only suspects?

Mass scanning
everyone is checked, all the time



Targeted
only suspects, with a judge’s order



The European Parliament supports the targeted model, while most member states want wider options. The negotiations have not settled this yet.

What we conclude

1. **The conflict is old.** From the Clipper Chip to Chat Control, each new technology has raised the same question again.

2. **Encryption moved it onto our devices.** Hacking, scanning and backdoors all reach into the phone itself.

3. **Targeted measures do better.** In our assessment they meet four of the five principles better than scanning everyone, and they still catch criminals.

4. **Infrastructure lasts.** What is built for surveillance stays available to future governments and to attackers.

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Thank you

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